

BEGINNER MATH

Mathematics

Master addition, subtraction, multiplication and division.

ADDITION SUBTRACTION MULTIPLY
DIVIDE ORDER OF OPERATIONS



IMPLOSE.GG

Mathematics

Course materials for addition, subtraction, multiplication, division, and mixed operations.

BEGINNER LEVEL

Read first. Then clear each quiz level with method, not guessing.

How To Use This Material

This course builds arithmetic fluency in five levels: addition, subtraction, multiplication, division, and a final mixed-operation boss battle. The goal is not to memorize the quiz answers. The goal is to understand the method well enough to solve the same kind of problem quickly and accurately.

For every calculation, write the operation clearly, solve one step at a time, and check whether the result is reasonable. Most mistakes in beginner arithmetic come from rushing, lining up digits incorrectly, or ignoring order of operations.

Course map

Level	Skill	What you must be able to do
1	Addition	Combine numbers, including two-digit and three-digit totals.
2	Subtraction	Find the difference and borrow correctly when needed.
3	Multiplication	Use times tables and repeated groups.
4	Division	Split into equal groups and connect division with multiplication.
5	Mixed operations	Choose the correct operation and follow order of operations.

A reliable solving routine

- Identify the operation: add, subtract, multiply, divide, or mixed.
- Estimate first. A rough answer helps catch impossible choices.
- Calculate carefully from right to left for column arithmetic.
- For mixed expressions, handle brackets first, then multiplication/division, then addition/subtraction.
- Check by using the opposite operation when possible.

Level 1 - Addition

Addition combines quantities. When numbers have more than one digit, line up place values: ones under ones, tens under tens, hundreds under hundreds.

- Use counting-on for small totals, such as adding a small number to a known starting value.
- Use place value for larger totals: add ones, then tens, then hundreds.
- If a column reaches 10 or more, carry 1 to the next column.
- A sum should be larger than each positive addend.

Worked examples

Example A: $34 + 21$

Ones: $4 + 1 = 5$.

Tens: 3 tens + 2 tens = 5 tens.

Answer: 55.

Example B: $67 + 18$

Ones: $7 + 8 = 15$, write 5 and carry 1.

Tens: $6 + 1 + \text{carried } 1 = 8$.

Answer: 85.

Common mistakes to avoid

- Do not add digits that are in different place-value columns.
- Do not forget a carried 1.
- Do not choose a smaller number when adding positive numbers.

Key takeaways

- Addition combines values into a larger total when all addends are positive.
- Place value keeps ones, tens, and hundreds in the correct columns.
- Estimation helps confirm that a final sum is reasonable.

Level 2 - Subtraction

Subtraction finds what remains or the difference between two numbers. Think of it as taking away or measuring the distance from one number to another.

- Line up place values before subtracting.
- If the top digit is smaller than the bottom digit, borrow 1 from the next place value.
- Subtraction can be checked with addition: difference + subtracted number = starting number.
- The answer should be smaller than the starting number when subtracting a positive number.

Worked examples

Example A: $63 - 27$

Ones: $3 - 7$ needs borrowing.

Borrow 1 ten: 63 becomes 5 tens and 13 ones.

Ones: $13 - 7 = 6$. Tens: $5 - 2 = 3$. Answer: 36.

Example B: $150 - 48$

Think 48 is close to 50.

$150 - 50 = 100$, then add back 2.

Answer: 102.

Common mistakes to avoid

- Do not subtract the smaller digit from the larger digit inside a column; borrow instead.
- Do not forget to reduce the column you borrowed from.
- Do not reverse subtraction order; starting value and amount removed are not interchangeable.

Key takeaways

- Subtraction finds a remaining amount or the distance between two values.
- Borrowing is a place-value trade, not a shortcut.
- Addition is the best way to check a subtraction result.

Level 3 - Multiplication

Multiplication is repeated addition of equal groups. For example, 4×8 means four groups of eight, or eight groups of four.

- Know core times tables from 1 to 12.
- Use commutativity: 8×9 equals 9×8 .

- Break hard facts into easier facts, such as splitting a number into tens and ones.
- Use nearby known facts and patterns, such as doubles, tens, and splitting numbers into parts.

Worked examples

Example A: 4×9

4 groups of 9 can be seen as 2 groups of 9 plus 2 groups of 9.

$$18 + 18 = 36.$$

Answer: 36.

Example B: 14×3

Break 14 into $10 + 4$.

$$10 \times 3 = 30 \text{ and } 4 \times 3 = 12.$$

$$30 + 12 = 42.$$

Common mistakes to avoid

- Do not treat multiplication like addition; equal groups grow faster than simple combining.
- Do not reverse digits after multiplying.
- Do not guess times-table facts without checking with a known pattern.

Key takeaways

- Multiplication represents equal groups, not random repeated counting.
- Known facts can be combined to solve unfamiliar facts.
- Breaking numbers apart makes larger multiplication easier.

Level 4 - Division

Division splits a total into equal groups or asks how many groups fit. It is the opposite of multiplication.

- Read a division expression as either splitting into equal groups or asking how many groups fit into the total.
- Use multiplication to check: if $81 \div 9 = 9$, then $9 \times 9 = 81$.
- For exact division in this course, there is no remainder.
- When choices are available, multiply each likely answer by the divisor to check.

Worked examples

Example A: $45 \div 5$

Ask: what number times 5 equals 45?

$$5 \times 9 = 45.$$

Answer: 9.

Example B: $132 \div 11$

Ask: 11 times what equals 132?

$$11 \times 10 = 110 \text{ and } 11 \times 2 = 22.$$

$$110 + 22 = 132, \text{ so the answer is } 12.$$

Common mistakes to avoid

- Do not divide the smaller number by the larger one unless the expression says so.
- Do not confuse division with subtraction; equal grouping is different from taking away once.

- Do not ignore the equal-groups idea; division is not random splitting.

Key takeaways

- Division and multiplication are inverse operations.
- Exact division means the total splits evenly into groups.
- A division result can be checked by multiplying it by the divisor.

Level 5 - Boss Battle: Mixed Operations

The final level mixes all four operations. The main skill is choosing the correct operation and using the correct order. In arithmetic, order matters.

Order of operations

Step	Rule	Example
1	Brackets first	$(18 - 6) / 4$ becomes $12 / 4$
2	Multiplication and division next, left to right	$8 + 3 \times 5$ becomes $8 + 15$
3	Addition and subtraction last, left to right	$8 + 15$ becomes 23

Worked examples

Example A: $5 \times 9 - 8$

Multiply first: $5 \times 9 = 45$.

Then subtract: $45 - 8 = 37$.

Example B: $8 \times 3 + 18 / 6$

Multiplication and division come before addition.

$8 \times 3 = 24$ and $18 / 6 = 3$.

Then add: $24 + 3 = 27$.

Example C: $(18 - 6) / 4$

Brackets first: $18 - 6 = 12$.

Then divide: $12 / 4 = 3$.

Boss Battle strategy

- Underline the operation symbols before calculating.
- Circle brackets if they appear.
- Do multiplication/division before addition/subtraction.
- After each step, rewrite the remaining expression so you do not lose track.
- Estimate the final answer to catch mistakes.

Common boss-level mistakes

- Solving strictly from left to right when multiplication should happen first.
- Forgetting brackets.
- Doing addition before division in expressions that contain both operation types.
- Copying the expression incorrectly after the first step.

Putting The Skills Together

Arithmetic becomes easier when each operation has a clear meaning. Addition combines, subtraction compares or removes, multiplication builds equal groups, and division separates into equal groups. Mixed expressions require a decision about sequence before calculation begins.

Mathematics vocabulary

Term	Meaning
Sum	The result of addition.
Difference	The result of subtraction.
Product	The result of multiplication.
Quotient	The result of division.
Place value	The value of a digit based on its position, such as ones, tens, or hundreds.
Expression	A mathematical phrase that contains numbers and operation symbols.

Mental math strategies

- Break numbers into friendly parts, especially tens and ones.
- Use nearby round numbers when subtracting, then adjust the result.
- Connect division facts to multiplication facts instead of treating them separately.
- Estimate first so the final result has a reasonable size.
- Keep one written line per step when an expression has more than one operation.

Study habits for quiz success

- Read the operation symbols before starting any calculation.
- Use place value for multi-digit addition and subtraction.
- Use multiplication facts to support both multiplication and division.
- Rewrite mixed expressions after each step so the remaining work stays clear.
- Check reasonableness with estimation before choosing or typing an answer.

The quiz should feel like an application of these methods. A learner who understands this material should be able to work through new arithmetic problems without needing a memorized answer list.